Exercise Sheet 26

Use the charts $z = v_1/v_0$, $w = v_0/v_1$ on $\mathbb{CP}^1$. A holomorphic section of $O(m)$ has entire coefficients f_0, f_1 with $f_1(w) = w^m f_0(1/w)$. For $m \geq 0$ it is a homogeneous polynomial $F(v_0, v_1)$ of degree m, with action $(\Pi_m(g)F)(v) = F(g^{-1}v)$. Exercises 26.1–26.5 are core; 26.6 is optional.

Exercise 26.1 (Changing bundle coordinates). Let $E = G \times_H V$ use $(gh, v) \sim (g, \sigma(h)v)$, and let $s_j = s_i h_{ij}$ be two local sections of $G \to G/H$.

(a) Derive the relation between the local coefficients $f_i = f \circ s_i$ and $f_j = f \circ s_j$ of a section.
(b) Replace each local section s_i by $s_i' = s_i a_i$, where $a_i : U_i \to H$ is smooth. Compute h_{ij}' and f_i'.
(c) Verify that the new coefficients have the new transition rule.

Exercise 26.2 (Regularity at infinity).

(a) For $m = 3$, decide whether $f_0(z) = 1 + 2z - z^3$, z^4, and e^z define global holomorphic sections. Give f_1 when it exists.
(b) Find all holomorphic sections for $m = 0$ and for $m = -2$.
(c) The expression $f_0(z) = z^3$ is allowed for $m = 2$ if meromorphic coefficients are permitted. Locate its pole in the other chart. Explain why it is excluded from $H^0(O(2))$.

Exercise 26.3 (A local group action). Let $g = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right) \in \mathrm{SU}(2)$.

(a) Let $f(z) = F(1, z)$ be the first-chart coefficient. Derive the formula

$$(\Pi_m(g)f)(z) = (\bar{a} + \bar{b}z)^m f((-b + az)/(\bar{a} + \bar{b}z)).$$

(b) For $m = 2$ and $g = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, compute the action on $1, z, z^2$ and its matrix in that basis.
(c) Verify that this matrix squares to the identity. Compare with $g^2 = -I$.

Exercise 26.4 (An invariant inner product). For homogeneous polynomials of degree m, put

$$\langle F, H \rangle = \int_{S^3} \overline{F(v)} H(v)\, d\mu(v),$$

where μ is the normalised round measure on the unit sphere in $\mathbb{C}^2$. It is the Haar integral of $\mathrm{SU}(2)$ transported by the first-column map, as in Lecture 5. You may use the moment formula proved in Exercise 5.2(b):

$$\int_{S^3} |v_0|^{2p} |v_1|^{2q}\, d\mu = \frac{p!q!}{(p+q+1)!}.$$

(a) Show that this is a positive definite $\mathrm{SU}(2)$-invariant inner product.
(b) Show that $F_k = v_0^{m-k} v_1^k$ are orthogonal and compute their squared norms. Use invariance under the two independent coordinate phases of S^3.
(c) Give an orthonormal basis when $m = 2$. Explain why the unscaled basis need not produce unitary arrays.

Exercise 26.5 (Duals and descent). Use the general convention $H^0(G/B, L_\lambda) = V_\lambda^*$.

(a) For SU(3), the dual of highest weight (a, b) has highest weight (b, a). Which bundle label realises the representation of highest weight $(2, 0)$?

(b) The centre element ζI of SU(3) acts on $V_{(a,b)}$ by ζ^{a+2b}, with $\zeta^3 = 1$. Find its action on the section space of $L_{(a,b)}$. When does this space descend to PSU(3)?

(c) Check the result for labels $(1, 0)$, $(1, 1)$ and $(3, 0)$.

Exercise 26.6 (Optional extension: multiplication of sections). For $m, n \geq 0$, multiply homogeneous polynomials to define

$$H^0(\mathcal{O}(m)) \otimes H^0(\mathcal{O}(n)) \longrightarrow H^0(\mathcal{O}(m+n)).$$

(a) Show that this map is well-defined on sections, is SU(2)-equivariant and is surjective.

(b) Compute the dimension of its kernel. For $m = n = 1$, identify a generator of the kernel.